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THE HINDU-ARABIC NUMERALS

BY

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AND

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Preface

So familiar are we with the numerals that bear the misleading name of Arabic, and so extensive is their use in Europe and the Americas, that it is difficult for us to realize that their general acceptance in the transactions of commerce is a matter of only the last four centuries, and that they are unknown to a very large part of the human race to-day.

It seems strange that such a labor-saving device should have struggled for nearly a thousand years after its system of place value was perfected before it replaced such crude notations as the one that the Roman conqueror made substantially universal in Europe. Such, however, is the case, and there is probably no one who has not at least some slight passing interest in the story of this struggle. To the mathematician and the student of civilization the interest is generally a deep one; to the teacher of the elements of knowledge the interest may be less marked, but nevertheless it is real; and even the business man who makes daily use of the curious symbols by which we express the numbers of commerce, cannot fail to have some appreciation for the story of the rise and progress of these tools of his trade.

This story has often been told in part, but it is a long time since any effort has been made to bring together the fragmentary narrations and to set forth the general problem of the origin and development of these numerals. In this little work we have attempted to state the history of these forms in small compass, to place before the student materials for the investigation of the problems involved, and to express as clearly as possible the results of the labors of scholars who have studied the subject in different parts of the world. We have had no theory to exploit, for the history of mathematics has seen too much of this tendency already, but as far as possible we have weighed the testimony and have set forth what seem to be the reasonable conclusions from the evidence at hand.

To facilitate the work of students an index has been prepared which we hope may be serviceable. In this the names of authors appear only when some use has been made of their opinions or when their works are first mentioned in full in a footnote.

If this work shall show more clearly the value of our number system, and shall make the study of mathematics seem more real to the teacher and student, and shall offer material for interesting some pupil more fully in his work with numbers, the authors will feel that the considerable labor involved in its preparation has not been in vain.

We desire to acknowledge our especial indebtedness to Professor Alexander Ziwet for reading all the proof, as well as for the digest of a Russian work, to Professor Clarence L. Meader for Sanskrit transliterations, and to Mr. Steven T. Byington for Arabic transliterations and the scheme of pronunciation of Oriental names, and also our indebtedness to other scholars in Oriental learning for information.

DAVID EUGENE SMITH
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Contents

Pronunciation of Oriental Names

(S) = in Sanskrit names and words; (A) = in Arabic names and words.

b, **d**, **f**, **g**, **h**, **j**, **l**, **m**, **n**, **p**, **sh (A)**, **t**, **th (A)**, **v**, **w**, **x**, **z**, as in English.

a, (S) like *u* in *but*: thus pandit, pronounced pundit. (A) like *a* in *ask* or in *man*.

a with macron, as in father.

c, (S) like *ch* in *church* (Italian *c* in *cento*).

d with dot below, **n with dot below**, **s with dot below**, **t with dot below**, (S) made with the tip of the tongue turned up and back into the dome of the palate.

d with dot below, **s with dot below**, **t with dot below**, **z with dot below**, (A) made with the tongue spread so that the sounds are produced largely against the side teeth.

d with dot below (A), like *th* in *this*.

e, (S) as in *they*. (A) as in *bed*.

g with dot below, (A) a voiced consonant formed below the vocal cords; its sound is compared by some to a *g*, by others to a guttural *r*; in Arabic words adopted into English it is represented by *gh* (e.g. ghoul), less often *r* (e.g. razzia).

h preceded by b, **c**, **t**, **etc**. does not form a single sound with these letters, but is a more or less distinct *h* sound following them; cf. the sounds in *abhor*, *boathook*, etc., or, more accurately for (S), the *bhoys* etc. of Irish brogue.

h (A) retains its consonant sound at the end of a word.

h with dot below, (A) an unvoiced consonant formed below the vocal cords; its sound is sometimes compared to German hard *ch*, and may be represented by an *h* as strong as possible.

i, as in *pin*. **i with macron**, as in *pique*.

k, as in *kick*.

kh, (A) the hard *ch* of Scotch *loch*, German *ach*, especially of German as pronounced by the Swiss.

m, **n**, (S) like French final *m* or *n*, nasalizing the preceding vowel.

n with dot below, see d with dot below. **n with dot above**, like *ng* in singing.

o, (S) as in *so*. (A) as in *obey*.

q, (A) like *k* (or *c*) in *cook*; further back in the mouth than in *kick*.

r, (S) English *r*, smooth and untrilled. (A) stronger. **r with dot below**, (S) used as vowel, as in *apron* when pronounced *aprn* and not *apera*; modern Hindus say *ri*, hence our amrita, Krishna, for a-mrta, Krsna.

s with acute, (S) English *sh* (German *sch*).

s, as in *same*. **s with dot below**, see d with dot below.

t, see d with dot below.

u, as in *put*. **u with macron**, as in *rule*.

y, as in *you*.

z, see d with dot below.

ayin / hamza ('), (A) a sound kindred to the spiritus lenis (that is, to our ears, the mere distinct separation of a vowel from the preceding sound, as at the beginning of a word in German) and to h with dot below. The ′ is a very distinct sound in Arabic, but is more nearly represented by the spiritus lenis than by any sound that we can produce without much special training. That is, it should be treated as silent, but the sounds that precede and follow it should not run together. In Arabic words adopted into English it is treated as silent, e.g. in Arab, amber, Caaba.

Accent: (S) as if Latin; (A) on the last syllable that contains a long vowel or a vowel followed by two consonants, except that a final long vowel is not ordinarily accented; if there is no long vowel nor two consecutive consonants, the accent falls on the first syllable. The words *al* and *ibn* are never accented.

1 Early Ideas of Their Origin

It has long been recognized that the common numerals used in daily life are of comparatively recent origin. The number of systems of notation employed before the Christian era was about the same as the number of written languages, and in some cases a single language had several systems. The Egyptians, for example, had three systems of writing, with a numerical notation for each; the Greeks had two well-defined sets of numerals, and the Roman symbols for number changed more or less from century to century. Even to-day the number of methods of expressing numerical concepts is much greater than one would believe before making a study of the subject, for the idea that our common numerals are universal is far from being correct.

It will be well, then, to think of the numerals that we still commonly call Arabic, as only one of many systems in use just before the Christian era. As it then existed the system was no better than many others, it was of late origin, it contained no zero, it was cumbersome and little used, and it had no particular promise. Not until centuries later did the system have any standing in the world of business and science; and had the place value which now characterizes it, and which requires a zero, been worked out in Greece, we might have been using Greek numerals to-day instead of the ones with which we are familiar.

Of the first number forms that the world used this is not the place to speak. Many of them are interesting, but none had much scientific value. In Europe the invention of notation was generally assigned to the eastern shores of the Mediterranean until the critical period of about a century ago,—sometimes to the Hebrews, sometimes to the Egyptians, but more often to the early trading Phoenicians.[1]

The idea that our common numerals are Arabic in origin is not an old one. The mediaeval and Renaissance writers generally recognized them as Indian, and many of them expressly stated that they were of Hindu origin. Others argued that they were probably invented by the Chaldeans or the Jews because they increased in value from right to left, an argument that would apply quite as well to the Roman and Greek systems, or to any other. It was, indeed, to the general idea of notation that many of these writers referred, as is evident from the words of England's earliest arithmetical textbook-maker, Robert Recorde (c. 1542):

> "In that thinge all men do agree, that the Chaldays, whiche fyrste inuented thys arte, did set these figures as thei set all their letters. for they wryte backwarde as you tearme it, and so doo they reade. And that may appeare in all Hebrewe, Chaldaye and Arabike bookes…where as the Greekes, Latines, and all nations of Europe, do wryte and reade from the lefte hand towarde the ryghte."[2]

Others, and among them such influential writers as Tartaglia in Italy and Kobel in Germany, asserted the Arabic origin of the numerals, while still others left the matter undecided or simply dismissed them as "barbaric." Of course the Arabs themselves never laid claim to the invention, always recognizing their indebtedness to the Hindus both for the numeral forms and for the distinguishing feature of place value.

Foremost among these writers was the great master of the golden age of Bagdad, one of the first of the Arab writers to collect the mathematical classics of both the East and the West, preserving them

[1]"Discipulus. Quis primus invenit numerum apud Hebraeos et Aegyptios ? Magister. Abraham primus invenit numerum apud Hebraeos, deinde Moses; et Abraham tradidit istam scientiam numeri ad Aegyptios, et docuit eos: deinde Josephus." [Bede, *De computo dialogus* (doubtfully assigned to him), Opera omnia, Paris, 1802, Vol. I, p. 650.] "Alii referunt ad Phoenices inventores arithmetices, propter eandem commerciorum caussam: Alii ad Indos: Ioannes de Sacrobosco, cujus sepulchrum est Lutetiae in comitio Maturinensi, refert ad Arabes." [Ramus, *Arithmeticae libri duo*, Basel, 1569, p. 112.]

[2]From the 1558 edition of *The Ground of Artes*, fol. C, 5. Similarly Bishop Tonstall writes: "Numerandi artem a Chaldaeis esse profectam: qui dum scribunt, a dextra incipiunt, et in leuam progreediuntur." [*De arte supputandi*, London, 1522, fol. B, 3.] Gemma Frisius, the great continental rival of Recorde, had the same idea: "Primum autem appellamus dexterum locum, eo quod haec ars vel a Chaldaeis, vel ab Hebraeis ortum habere credatur, qui etiam eo ordine scribunt"; but this refers more evidently to the Arabic numerals. [*Arithmeticae practicae methodus facilis*, Antwerp, 1540, fol. 4 of the 1563 ed.]

and finally passing them on to awakening Europe. This man was Mohammed the Son of Moses, from Khowarezm, or, more after the manner of the Arab, **Mohammed ibn Musa al-Khowarazmi**,[3] a man of great learning and one to whom the world is much indebted for its present knowledge of algebra and of arithmetic.

Of him there will often be occasion to speak; and in the arithmetic which he wrote, and of which Adelhard of Bath (c. 1130) may have made the translation or paraphrase, he stated distinctly that the numerals were due to the Hindus. This is as plainly asserted by later Arab writers, even to the present day. Indeed the phrase *'ilm hindi*, "Indian science," is used by them for arithmetic, as also the adjective *hindi* alone.

Probably the most striking testimony from Arabic sources is that given by the Arabic traveler and scholar **Mohammed ibn Ahmed**, **Abu 'l-Rihan al-Biruni** (973–1048), who spent many years in Hindustan. He wrote a large work on India, the "Book of the Ciphers," unfortunately lost, which treated doubtless of the Hindu art of calculating, and was the author of numerous other works. Al-Biruni was a man of unusual attainments, being versed in Arabic, Persian, Sanskrit, Hebrew, and Syriac, as well as in astronomy, chronology, and mathematics. In his work on India he gives detailed information concerning the language and customs of the people of that country, and states explicitly that the Hindus of his time did not use the letters of their alphabet for numerical notation, as the Arabs did. He also states that the numeral signs called *ahka*[4] had different shapes in various parts of India, as was the case with the letters. In his *Chronology of Ancient Nations* he gives the sum of a geometric progression and shows how, in order to avoid any possibility of error, the number may be expressed in three different systems: with Indian symbols, in sexagesimal notation, and by an alphabet system which will be touched upon later. He also speaks of "179, 876, 755, expressed in Indian ciphers," thus again attributing these forms to Hindu sources.

Preceding Al-Biruni there was another Arabic writer of the tenth century, Motahhar ibn Tahir, author of the Book of the Creation and of History, who gave as a curiosity, in Indian (Nagari) symbols, a large number asserted by the people of India to represent the duration of the world.

Mention should also be made of a widely-traveled student, **Al-Mas'udi** (885?–956), whose journeys carried him from Bagdad to Persia, India, Ceylon, and even across the China sea. He states that the wise men of India, assembled by the king, composed the *Sindhind*. Further on he states, upon the authority of the historian Mohammed ibn 'Ali 'Abdi, that by order of Al-Mansur many works of science and astrology were translated into Arabic, notably the Sindhind (Siddhanta). Concerning the meaning and spelling of this name there is considerable diversity of opinion. Colebrooke first pointed out the connection between Siddhanta and Sindhind. He ascribes to the word the meaning "the revolving ages."

This Sindhind is the book, says Mas'udi, which gives all that the Hindus know of the spheres, the stars, arithmetic, and the other branches of science. He mentions also Al-Khowarazmi and Habash as translators of the tables of the Sindhind. Al-Biruni refers to two other translations from a work furnished by a Hindu who came to Bagdad as a member of the political mission which Sindh sent to the caliph Al-Mansur, in the year of the Hejira 154 (a.d. 771).

The oldest work, in any sense complete, on the history of Arabic literature and history is the *Kitab al-Fihrist*, written in the year 987 a.d., by Ibn Abi Ya'qub al-Nadim. It is of fundamental importance for the history of Arabic culture. Of the ten chief divisions of the work, the seventh demands attention in this discussion for the reason that its second subdivision treats of mathematicians and astronomers.

The first of the Arabic writers mentioned is Al-Kindi (800–870 A.D.), who wrote five books on arithmetic and four books on the use of the Indian method of reckoning. Sened ibn 'Ali, the Jew, who was converted to Islam under the caliph Al-Mamun, is also given as the author of a work on the Hindu

[3]His full name was Abu 'Abdallah Mohammed ibn Musa al-Khowarazmi. He was born in Khowarezm, "the lowlands," the country about the present Khiva and bordering on the Oxus, and lived at Bagdad under the caliph al-Mamun. He died probably between 220 and 230 of the Mohammedan era, that is, between 835 and 845 a.d., although some put the date as early as 812.

[4]The Hindu name for the symbols of the decimal place system.

method of reckoning. To Al-Sufi, who died in 986 a.d., is also credited a large work on the same subject, and similar treatises by other writers are mentioned. We are therefore forced to the conclusion that the Arabs from the early ninth century on fully recognized the Hindu origin of the new numerals.

Leonard of Pisa, of whom we shall speak at length in the chapter on the Introduction of the Numerals into Europe, wrote his *Liber Abbaci* in 1202. In this work he refers frequently to the nine Indian figures, thus showing again the general consensus of opinion in the Middle Ages that the numerals were of Hindu origin. Some interest also attaches to the oldest documents on arithmetic in our own language. One of the earliest treatises on algorism is a commentary on a set of verses called the *Carmen de Algorismo*, written by Alexander de Villa Dei (Alexandre de Ville-Dieu), a Minorite monk of about 1240 a.d. The text of the first few lines is as follows:

> *Haec algorismus ars praesens dicitur, in qua*
> *Talibus Indorum fruimur bis quinque figuris.*

"This boke is called the boke of algorim or augrym after lewder use. And this boke tretys of the Craft of Nombryng, the quych crafte is called also Algorym. Ther was a kyng of Inde the quich heyth Algor and he made this craft. Algorisms, in the quych we use teen figurys of Inde."

2 Early Hindu Forms with No Place Value

While it is generally conceded that the scientific development of astronomy among the Hindus towards the beginning of the Christian era rested upon Greek or Chinese sources, yet their ancient literature testifies to a high state of civilization, and to a considerable advance in sciences, in philosophy, and along literary lines, long before the golden age of Greece. From the earliest times even up to the present day the Hindu has been wont to put his thought into rhythmic form. The first of this poetry—it well deserves this name, being also worthy from a metaphysical point of view—consists of the Vedas, hymns of praise and poems of worship, collected during the Vedic period which dates from approximately 2000 B.C. to 1400 B.C.

Following this work, or possibly contemporary with it, is the Brahmanic literature, which is partly ritualistic (the Brahmanas), and partly philosophical (the Upanishads). Our especial interest is in the Sutras, versified abridgments of the ritual and of ceremonial rules, which contain considerable geometric material used in connection with altar construction, and also numerous examples of rational numbers the sum of whose squares is also a square, i.e. Pythagorean numbers, although this was long before Pythagoras lived.

Whitney places the whole of the Veda literature, including the Vedas, the Brahmanas, and the Sutras, between 1500 B.C. and 800 B.C., thus agreeing with Birk who holds that the knowledge of the Pythagorean theorem revealed in the Sutras goes back to the eighth century B.C. The importance of the Sutras as showing an independent origin of Hindu geometry, contrary to the opinion long held by Cantor of a Greek origin, has been repeatedly emphasized in recent literature, especially since the appearance of the important work of Von Schroeder.

Further fundamental mathematical notions such as the conception of irrationals and the use of gnomons, as well as the philosophical doctrine of the transmigration of souls,—all of these having long been attributed to the Greeks,—are shown in these works to be native to India. Although this discussion does not bear directly upon the origin of our numerals, yet it is highly pertinent as showing the aptitude of the Hindu for mathematical and mental work, a fact further attested by the independent development of the drama and of epic and lyric poetry.

It should be stated definitely at the outset, however, that we are not at all sure that the most ancient forms of the numerals commonly known as Arabic had their origin in India. As will presently be seen, their forms may have been suggested by those used in Egypt, or in Eastern Persia, or in China, or on the plains of Mesopotamia. We are quite in the dark as to these early steps; but as to their development in

India, the approximate period of the rise of their essential feature of place value, their introduction into the Arab civilization, and their spread to the West, we have more or less definite information. When, therefore, we consider the rise of the numerals in the land of the Sindhu, it must be understood that it is only the large movement that is meant, and that there must further be considered the numerous possible sources outside of India itself and long anterior to the first prominent appearance of the number symbols.

No one attempts to examine any detail in the history of ancient India without being struck with the great dearth of reliable material. So little sympathy have the people with any save those of their own caste that a general literature is wholly lacking, and it is only in the observations of strangers that any all-round view of scientific progress is to be found.

There is evidence that primary schools existed in earliest times, and of the seventy-two recognized sciences writing and arithmetic were the most prized. In the Vedic period, say from 2000 to 1400 B.C., there was the same attention to astronomy that was found in the earlier civilizations of Babylon, China, and Egypt, a fact attested by the Vedas themselves. Such advance in science presupposes a fair knowledge of calculation, but of the manner of calculating we are quite ignorant and probably always shall be.

One of the Buddhist sacred books, the *Lalitavistara*, relates that when the Bodhisattva was of age to marry, the father of Gopa, his intended bride, demanded an examination of the five hundred suitors, the subjects including arithmetic, writing, the lute, and archery. Having vanquished his rivals in all else, he is matched against Arjuna the great arithmetician and is asked to express numbers greater than 100 kotis.[5] In reply he gave a scheme of number names as high as 10^{53}, adding that he could proceed as far as 10^{421}, all of which suggests the system of Archimedes and the unsettled question of the indebtedness of the West to the East in the realm of ancient mathematics.

Sir Edwin Arnold, in *The Light of Asia*, does not mention this part of the contest, but he speaks of Buddha's training at the hands of the learned Visvamitra:

> And Viswamitra said, 'It is enough,
> Let us to numbers. After me repeat
> Your numeration till we reach the lakh,[6]
> One, two, three, four, to ten, and then by tens
> To hundreds, thousands.' After him the child
> Named digits, decads, centuries, nor paused,
> The round lakh reached, but softly murmured on,
> Then comes the koti, nahut, ninahut,
> Khamba, viskhamba, abab, attata,
> To kumuds, gundhikas, and utpalas,
> By pundarikas into padumas,
> Which last is how you count the utmost grains
> Of Hastagiri ground to finest dust;
> But beyond that a numeration is,
> The Katha, used to count the stars of night,
> The Koti-Katha, for the ocean drops;
> Ingga, the calculus of circulars;
> Sarvanikchepa, by the which you deal
> With all the sands of Gunga, till we come
> To Antah-Kalpas, where the unit is
> The sands of the ten crore Gungas. If one seeks

[5] I.e. $100 \cdot 10^7$.

[6] I.e. to 100,000. The lakh is still the common large unit in India, like the myriad in ancient Greece and the million in the West.

More comprehensive scale, th' arithmic mounts
By the Asankya, which is the tale
Of all the drops that in ten thousand years
Would fall on all the worlds by daily rain;
Thence unto Maha Kalpas, by the which
The gods compute their future and their past.'

Thereupon Visvamitra Acarya[7] expresses his approval of the task, and asks to hear the "measure of the earth" as far as yojana, the longest measure bearing name. This given, Buddha adds:

'And master! if it please,
I shall recite how many sun-motes lie
From end to end within a yojana.'
Thereat, with instant skill, the little prince
Pronounced the total of the atoms true.
But Viswamitra heard it on his face
Prostrate before the boy;
'Art Teacher of thy teachers—thou, not I,
For thou,' he cried, 'Art Guru.'

It is needless to say that this is far from being history. And yet it puts in charming rhythm only what the ancient Lalitavistara relates of the number-series of the Buddha's time. While it extends beyond all reason, nevertheless it reveals a condition that would have been impossible unless arithmetic had attained a considerable degree of advancement.

To this pre-Christian period belong also the Veddngas, or "limbs for supporting the Veda," part of that great branch of Hindu literature known as Smriti (recollection), that which was to be handed down by tradition. Of these the sixth is known as Jyotish (astronomy), a short treatise of only thirty-six verses, written not earlier than 300 B.C., and affording us some knowledge of the extent of number work in that period.

The Hindus also speak of eighteen ancient Siddhantas or astronomical works, which, though mostly lost, confirm this evidence. As to authentic histories, however, there exist in India none relating to the period before the Mohammedan era (622 a.d.). About all that we know of the earlier civilization is what we glean from the two great epics, the Mahabharata and the Ramayana, from coins, and from a few inscriptions.

It is with this unsatisfactory material, then, that we have to deal in searching for the early history of the Hindu-Arabic numerals, and the fact that many unsolved problems exist and will continue to exist is no longer strange when we consider the conditions. It is rather surprising that so much has been discovered within a century, than that we are so uncertain as to origins and dates and the early spread of the system.

The early Hindu numerals may be classified into three great groups, (1) the Kharosthi, (2) the Brahmi, and (3) the word and letter forms; and these will be considered in order.

The Kharosthi numerals are found in inscriptions formerly known as Bactrian, Indo-Bactrian, and Aryan, and appearing in ancient Gandhara, now eastern Afghanistan and northern Punjab. The alphabet of the language is found in inscriptions dating from the fourth century B.C. to the third century A.D., and from the fact that the words are written from right to left it is assumed to be of Semitic origin. No numerals, however, have been found in the earliest of these inscriptions, number-names probably having been written out in words as was the custom with many ancient peoples.

Not until the time of the powerful King Asoka, in the third century B.C., do numerals appear in any inscriptions thus far discovered; and then only in the primitive form of marks, quite as they would be

[7] I.e. the Wise.

found in Egypt, Greece, Rome, or in various other parts of the world. These Asoka inscriptions, some thirty in all, are found in widely separated parts of India, often on columns, and are in the various vernaculars that were familiar to the people. Two are in the Kharosthi characters, and the rest in some form of Brahmi. In the Kharosthi inscriptions only four numerals have been found, and these are merely vertical marks for one, two, four, and five.

In the so-called Saka inscriptions, possibly of the first century B.C., more numerals are found, and in more highly developed form, the right-to-left system appearing, together with evidences of three different scales of counting,—four, ten, and twenty.

There are several noteworthy points to be observed in studying this system. In the first place, it is probably not as early as that shown in the Nana Ghat forms hereafter given, although the inscriptions themselves at Nana Ghat are later than those of the Asoka period. The four is to this system what the X was to the Roman, probably a canceling of three marks as a workman does to-day for five, or a laying of one stick across three others. The ten has never been satisfactorily explained. It is similar to the A of the Kharosthi alphabet, but we have no knowledge as to why it was chosen. The twenty is evidently a ligature of two tens, and this in turn suggested a kind of radix, so that ninety was probably written in a way reminding one of the *quatre-vingt-dix* of the French.

This system has many points of similarity with the Nabatean numerals in use in the first centuries of the Christian era. The cross is here used for four, and the Kharosthi form is employed for twenty. In addition to this there is a trace of an analogous use of a scale of twenty. While the symbol for 100 is quite different, the method of forming the other hundreds is the same. The correspondence seems to be too marked to be wholly accidental.

It is not in the Kharosthi numerals, therefore, that we can hope to find the origin of those used by us, and we turn to the second of the Indian types, the Brahmi characters. The alphabet attributed to Brahma is the oldest of the several known in India, and was used from the earliest historic times.

The following numerals are, as far as known, the only ones to appear in the Asoka edicts. These fragments from the third century B.C., crude and unsatisfactory as they are, are the undoubted early forms from which our present system developed. They next appear in the second century B.C. in some inscriptions in the cave on the top of the Nana Ghat hill, about seventy-five miles from Poona in central India.

The cave was made as a resting-place for travelers ascending the hill, which lies on the road from Kalyana to Junar. It seems to have been cut out by a descendant of King Satavahana, for inside the wall opposite the entrance are representations of the members of his family, much defaced, but with the names still legible. It would seem that the excavation was made by order of a king named Vedisiri, and "the inscription contains a list of gifts made on the occasion of the performance of several yagnas or religious sacrifices," and numerals are to be seen in no less than thirty places.

There is considerable dispute as to what numerals are really found in these inscriptions, owing to the difficulty of deciphering them; but the following, which have been copied from a rubbing, are probably number forms.

The next very noteworthy evidence of the numerals, and this quite complete as will be seen, is found in certain other cave inscriptions dating back to the first or second century A.D. In these, the Nasik cave inscriptions, the forms are as follows.

From this time on, until the decimal system finally adopted the first nine characters and replaced the rest of the Brahmi notation by adding the zero, the progress of these forms is well marked.

With respect to these numerals it should first be noted that no zero appears in the table, and as a matter of fact none existed in any of the cases cited. It was therefore impossible to have any place value, and the numbers like twenty, thirty, and other multiples of ten, one hundred, and so on, required separate symbols except where they were written out in words. The ancient Hindus had no less than twenty of these symbols, a number that was afterward greatly increased.

We have thus far spoken of the Kharosthi and Brahmi numerals, and it remains to mention the third type, the word and letter forms. These are, however, so closely connected with the perfecting of the system by the invention of the zero that they are more appropriately considered in the next chapter, particularly as they have little relation to the problem of the origin of the forms known as the Arabic.

Having now examined types of the early forms it is appropriate to turn our attention to the question of their origin. As to the first three there is no question. The I or — is simply one stroke, or one stick laid down by the computer. The II represents two strokes or two sticks, and so for the III and IV. From some primitive form came the two of Egypt, of Rome, of early Greece, and of various other civilizations.

From some primitive form — came the Chinese symbol, which is practically identical with the symbols found commonly in India from 150 B.C. to 700 A.D. In the cursive form it becomes *z*, and this was frequently used for two in Germany until the 18th century. It finally went into the modern form 2, and the = in the same way became our 3.

There is, however, considerable ground for interesting speculation with respect to these first three numerals. The earliest Hindu forms were perpendicular. In the Nana Ghat inscriptions they are vertical. But long before either the Asoka or the Nana Ghat inscriptions the Chinese were using the horizontal forms for the first three numerals, but a vertical arrangement for four. Now where did China get these forms? Surely not from India, for she had them, as her monuments and literature show, long before the Hindus knew them.

The tradition is that China brought her civilization around the north of Tibet, from Mongolia, the primitive habitat being Mesopotamia, or possibly the oases of Turkestan. Now what numerals did Mesopotamia use? The Babylonian system, simple in its general principles but very complicated in many of its details, is now well known. In particular, one, two, and three were represented by vertical arrow-heads. Why, then, did the Chinese write theirs horizontally? The problem now takes a new interest when we find that these Babylonian forms were not the primitive ones of this region, but that the early Sumerian forms were horizontal.

What interpretation shall be given to these facts? Shall we say that it was mere accident that one people wrote “one” vertically and that another wrote it horizontally? This may be the case; but it may also be the case that the tribal migrations that ended in the Mongol invasion of China started from the Euphrates while yet the Sumerian civilization was prominent, or from some common source in Turkestan, and that they carried to the East the primitive numerals of their ancient home, the first three, these being all that the people as a whole knew or needed.

As to the numerals above three there have been very many conjectures. The figure one of the Demotic looks like the one of the Sanskrit, the two (reversed) like that of the Arabic, the four has some resemblance to that in the Nasik caves, the five (reversed) to that on the Ksatrapa coins, the nine to that of the Kusana inscriptions, and other points of similarity have been imagined.

More absurd is the hypothesis of a Greek origin, supposedly supported by derivation of the current symbols from the first nine letters of the Greek alphabet. This difficult feat is accomplished by twisting some of the letters, cutting off, adding on, and effecting other changes to make the letters fit the theory. This peculiar theory was first set up by Dasypodius (Conrad Rauhfuss), and was later elaborated by Huet.

Out of all these conflicting theories, and from all the resemblances seen or imagined between the numerals of the West and those of the East, what conclusions are we prepared to draw as the evidence now stands? Probably none that is satisfactory. Indeed, upon the evidence at hand we might properly feel that everything points to the numerals as being substantially indigenous to India. And why should this not be the case? If the king Srong-tsan-Gampo (639 a.d.), the founder of Lhasa, could have set about to devise a new alphabet for Tibet, and if the Siamese, and the Singhalese, and the Burmese, and other peoples in the East, could have created alphabets of their own, why should not the numerals also have been fashioned by some temple school, or some king, or some merchant guild?

We may summarize this chapter by saying that no one knows what suggested certain of the early numeral forms used in India. The origin of some is evident, but the origin of others will probably never be known. There is no reason why they should not have been invented by some priest or teacher or guild, by the order of some king, or as part of the mysticism of some temple. Whatever the origin, they were no better than scores of other ancient systems and no better than the present Chinese system when written without the zero, and there would never have been any chance of their triumphal progress westward had it not been for this relatively new symbol.

3 Later Hindu Forms, with a Place Value

Before speaking of the perfected Hindu numerals with the zero and the place value, it is necessary to consider the third system mentioned above—the word and letter forms. The use of words with place value began at least as early as the 6th century of the Christian era. In many of the manuals of astronomy and mathematics, and often in other works in mentioning dates, numbers are represented by the names of certain objects or ideas.

For example, zero is represented by "heaven-space" or "the void" (*ambara* or *akasa*); one by "moon" (*sasi* or *indu*), "earth" (*bhu*), "beginning" (*adi*), "Brahma," or, in general, by anything markedly unique; two by "eyes" (*nayana*) or "hands" (*kara*), "the twins" or "Yama"; four by "oceans," five by "senses" or "arrows" (the five arrows of Kamadeva); six by "seasons" or "flavors"; seven by "mountain" (*ago*), and so on. These names, accommodating themselves to the verse in which scientific works were written, had the additional advantage of not admitting, as did the figures, easy alteration, since any change would tend to disturb the meter.

As an example of this system, the date "Saka Samvat, 867" (a.d. 945 or 940), is given by "*giri-rasa-vasu*" meaning "mountains" (seven), "the flavors" (six), and the gods of which there were eight. In reading the date these are read from right to left. The period of invention of this system is uncertain. The first trace seems to be in the Srautasutra of Katyayana and Latyayana. It was certainly known to Varaha-Mihira (d. 587), for he used it in the *Brhat-Samhita*.

Mention should also be made, in this connection, of a curious system of alphabetic numerals that sprang up in southern India. In this we have the numerals represented by the letters as given in the following table.

By this plan a numeral might be represented by any one of several letters, and thus it could the more easily be formed into a word for mnemonic purposes. For example, the word *kha gont yan me sa ma pa* has the value 1,565,132, reading from right to left. This, the oldest specimen (1184 a.d.) known of this notation, is given in a commentary on the Rigveda, representing the number of days that had elapsed from the beginning of the Kaliyuga.

A second system of this kind is still used in the pagination of manuscripts in Ceylon, Siam, and Burma, having also had its rise in southern India. In this the thirty-four consonants when followed by *a* (as *ka*, *la*) designate the numbers 1–34; by *a* (as *ka*, *la*), those from 35 to 68; by *i* (*ki*, *li*), those from 69 to 102, inclusive; and so on.

As already stated, however, the Hindu system as thus far described was no improvement upon many others of the ancients, such as those used by the Greeks and the Hebrews. Having no zero, it was impracticable to designate the tens, hundreds, and other units of higher order by the same symbols used for the units from one to nine. In other words, there was no possibility of place value without some further improvement. So the Nana Ghat symbols required the writing of "thousand seven twenty-four" about like 7, tw, 4 in modern symbols, instead of 7024, in which the seven of the thousands, the two of the tens (concealed in the word twenty, being originally "twain of tens," the *-ty* signifying ten), and the four of the units are given as spoken and the order of the unit (tens, hundreds, etc.) is given by the place. To complete the system only the zero was needed; but it was probably eight centuries after the Nana Ghat inscriptions were cut, before this important symbol appeared; and not until a considerably later period did it become well known.

To enable us to appreciate the force of this argument a large number, 8,443,682,155, may be considered as the Hindus wrote and read it, and then, by way of contrast, as the Greeks and Arabs would have read it.

> **Modern American reading**: 8 billion, 443 million, 682 thousand, 155.
>
> **Hindu**: 8 padmas, 4 vyarbudas, 4 kotis, 3 prayutas, 6 laksas, 8 ayutas, 2 sahasra, 1 sata, 5 dasan, 5.
>
> **Arabic and early German**: eight thousand thousand thousand and four hundred thousand thousand and forty-three thousand thousand, and six hundred thousand and eighty-two thousand and one hundred fifty-five (or five and fifty).
>
> **Greek**: eighty-four myriads of myriads and four thousand three hundred sixty-eight myriads and two thousand and one hundred fifty-five.

As Woepcke pointed out, the reading of numbers of this kind shows that the notation adopted by the Hindus tended to bring out the place idea. No other language than the Sanskrit has made such consistent application, in numeration, of the decimal system of numbers. The introduction of myriads as in the Greek, and thousands as in Arabic and in modern numeration, is really a step away from a decimal scheme.

When the *ankapalli*, the decimal-place system of writing numbers, was perfected, the tenth symbol was called the *sunyabindu*, generally shortened to *sunya* (the void). Broekhaus has well said that if there was any invention for which the Hindus, by all their philosophy and religion, were well fitted, it was the invention of a symbol for zero. This making of nothingness the crux of a tremendous achievement was a step in complete harmony with the genius of the Hindu.

It is generally thought that this *sunya* as a symbol was not used before about 500 A.D., although some writers have placed it earlier. Since Aryabhata gives our common method of extracting roots, it would seem that he may have known a decimal notation, although he did not use the characters from which our numerals are derived. Moreover, he frequently speaks of the *void*.[8] If he refers to a symbol this would put the zero as far back as 500 a.d., but of course he may have referred merely to the concept of nothingness.

A little later, but also in the sixth century, Varaha-Mihira wrote a work entitled *Brhat Samhita* in which he frequently uses *sunya* in speaking of numerals, so that it has been thought that he was referring to a definite symbol. This, of course, would add to the probability that Aryabhata was doing the same. It should also be mentioned as a matter of interest, and somewhat related to the question at issue, that Varaha-Mihira used the word-system with place value as explained above.

The first kind of alphabetic numerals and also the word-system (in both of which the place value is used) are plays upon, or variations of, position arithmetic, which would be most likely to occur in the country of its origin.

At the opening of the next century (c. 620 a.d.) Bana wrote of Subandhu's *Vasavadatta* as a celebrated work, and mentioned that the stars dotting the sky are here compared with zeros, these being points as in the modern Arabic system. On the other hand, a strong argument against any Hindu knowledge of the symbol zero at this time is the fact that about 700 A.D. the Arabs overran the province of Sind and thus had an opportunity of knowing the common methods used there for writing numbers. And yet, when they received the complete system in 776 they looked upon it as something new. Such evidence is not conclusive, but it tends to show that the complete system was probably not in common use in India at the beginning of the eighth century.

On the other hand, we must bear in mind the fact that a traveler in Germany in the year 1700 would probably have heard or seen nothing of decimal fractions, although these were perfected a century before that date. The elite of the mathematicians may have known the zero even in Aryabhata's time,

[8] Using *kha*, a synonym of *sunya*.

while the merchants and the common people may not have grasped the significance of the novelty until a long time after. On the whole, the evidence seems to point to the west coast of India as the region where the complete system was first seen.

Concerning the earliest epigraphical instances of the use of the nine symbols, plus the zero, with place value, there is some question. Colebrooke in 1807 warned against the possibility of forgery in many of the ancient copper-plate land grants. On this account Fleet, in the Indian Antiquary, discusses at length this phase of the work of the epigraphists in India, holding that many of these forgeries were made about the end of the eleventh century.

Buhler gives the copper-plate Gurjara inscription of Cedi-samvat 346 (595 a.d.) as the oldest epigraphical use of the numerals "in which the symbols correspond to the alphabet numerals of the period and the place." Vincent A. Smith quotes a stone inscription of 815 A.D., dated Samvat 872. So F. Kielhorn in the *Epigraphia Indica* gives a Pathari pillar inscription of Parabala, dated Vikrama-samvat 917, which corresponds to 861 A.D.

That there should be any inscriptions of date as early even as 750 A.D., would tend to show that the system was at least a century older. As will be shown in the further development, it was more than two centuries after the introduction of the numerals into Europe that they appeared there upon coins and inscriptions.

While Thibaut does not consider it necessary to quote any specific instances of the use of the numerals, he states that traces are found from 590 a.d. on. "That the system now in use by all civilized nations is of Hindu origin cannot be doubted; no other nation has any claim upon its discovery, especially since the references to the origin of the system which are found in the nations of western Asia point unanimously towards India."

The testimony and opinions of men like Buhler, Kielhorn, V. A. Smith, Bhandarkar, and Thibaut are entitled to the most serious consideration. As authorities on ancient Indian epigraphy no others rank higher. Their work is accepted by Indian scholars the world over, and their united judgment as to the rise of the system with a place value—that it took place in India as early as the sixth century a.d.—must stand unless new evidence of great weight can be submitted to the contrary.

4 The Symbol Zero

What has been said of the improved Hindu system with a place value does not touch directly the origin of a symbol for zero, although it assumes that such a symbol exists. The importance of such a sign, the fact that it is a prerequisite to a place-value system, and the further fact that without it the Hindu-Arabic numerals would never have dominated the computation system of the western world, make it proper to devote a chapter to its origin and history.

It was some centuries after the primitive Brahmi and Kharosthi numerals had made their appearance in India that the zero first appeared there, although such a character was used by the Babylonians in the centuries immediately preceding the Christian era. The symbol is known to us from their tablets, and apparently it was not used in calculation. Nor does it always occur when units of any order are lacking; thus 180 is written with the meaning three sixties and no units, since 181 immediately following is three sixties and one unit.

The earliest undoubted occurrence of a zero in India is an inscription at Gwalior, dated Samvat 933 (876 A.D.). Where 50 garlands are mentioned (line 20), 50 is written. The Bakshali Manuscript probably antedates this, using the point or dot as a zero symbol. Bayley mentions a grant of Jaika Rashtrakuta of Bharuj, found at Okamandel, of date 738 A.D., which contains a zero, and also a coin with indistinct Gupta date 707 (897 a.d.), but the reliability of Bayley's work is questioned.

Aside from its appearance in early inscriptions, there is still another indication of the Hindu origin of the symbol in the special treatment of the concept zero in the early works on arithmetic. Brahmagupta, who lived in Ujjain, the center of Indian astronomy, in the early part of the seventh century, gives in his arithmetic a distinct treatment of the properties of zero. He does not discuss a symbol, but he shows by his treatment that in some way zero had acquired a special significance not found in the Greek or other ancient arithmetics. A still more scientific treatment is given by Bhaskara, although in one place he permits himself an unallowed liberty in dividing by zero. The most recently discovered work of ancient Indian mathematical lore, the *Ganita-Sara-Sangraha* of Mahaviracarya (c. 830 a.d.), while it does not use the numerals with place value, has a similar discussion of the calculation with zero.

What suggested the form for the zero is, of course, purely a matter of conjecture. The dot, which the Hindus used to fill up lacunae in their manuscripts, much as we indicate a break in a sentence, would have been a more natural symbol; and this is the one which the Hindus first used and which most Arabs use to-day. There was also used for this purpose a cross, like our X, and this is occasionally found as a zero symbol.

The small circle was possibly suggested by the spurred circle which was used for ten. It has also been thought that the omicron used by Ptolemy in his Almagest, to mark accidental blanks in the sexagesimal system which he employed, may have influenced the Indian writers. This symbol was used quite generally in Europe and Asia, and the Arabic astronomer Al-Battani (died 929 a.d.) used a similar symbol in connection with the alphabetic system of numerals.

Although the dot was used at first in India, as noted above, the small circle later replaced it and continues in use to this day. The Arabs, however, did not adopt the circle, since it bore some resemblance to the letter which expressed the number five in the alphabet system. The earliest Arabic zero known is the dot, used in a manuscript of 873 a.d. Sometimes both the dot and the circle are used in the same work, having the same meaning.

The Name of Zero

The name of this all-important symbol also demands some attention, especially as we are even yet quite undecided as to what to call it. We speak of it to-day as zero, naught, and even cipher; the telephone operator often calls it O, and the illiterate or careless person calls it aught. In view of all this uncertainty we may well inquire what it has been called in the past.

As already stated, the Hindus called it *sunya*, "void." This passed over into the Arabic as *as-sifr* or *sifr*. When Leonard of Pisa (1202) wrote upon the Hindu numerals he spoke of this character as *zephirum*. Maximus Planudes (1330), writing under both the Greek and the Arabic influence, called it *tziphra*.

Of course the English *cipher*, French *chiffre*, is derived from the same Arabic word, *as-sifr*, but in several languages it has come to mean the numeral figures in general. A trace of this appears in our word *ciphering*, meaning figuring or computing.

From *as-sifr* has come *zephyr*, *cipher*, and finally the abridged form *zero*. The earliest printed work in which is found this final form appears to be Calandri's arithmetic of 1491, while in manuscript it appears at least as early as the middle of the fourteenth century. It also appears in a work, *Le Kadran des marchans*, by Jehan Certain, written in 1485. This word soon became fairly well known in Spain and France.

The medieval writers also spoke of it as the *sipos*, and occasionally as the *galgal* or "wheel," *circulus*, *nota*, *theca*,[9] *omicron* (the Greek o), being sometimes written *o* or ϕ to distinguish it from the letter *o*. It also went by the name *null* or *nulla*, and very commonly by the name *cipher*.

5 The Question of the Introduction of the Numerals into Europe by Boethius

Just as we were quite uncertain as to the origin of the numeral forms, so too are we uncertain as to the time and place of their introduction into Europe. There are two general theories as to this

[9] Long supposed to be from its resemblance to the Greek theta, but explained by Petrus de Dacia as being derived from the name of the iron used to brand thieves and robbers with a circular mark placed on the forehead or on the cheek.

introduction. The first is that they were carried by the Moors to Spain in the eighth or ninth century, and thence were transmitted to Christian Europe, a theory which will be considered later. The second, advanced by Woepcke, is that they were not brought to Spain by the Moors, but that they were already in Spain when the Arabs arrived there, having reached the West through the Neo-Pythagoreans.

There are two facts to support this second theory: (1) the forms of these numerals are characteristic, differing materially from those which were brought by Leonardo of Pisa from Northern Africa early in the thirteenth century (before 1202 a.d.); (2) they are essentially those which tradition has so persistently assigned to Boethius (c. 500 A.D.), and which he would naturally have received, if at all, from these same Neo-Pythagoreans or from the sources from which they derived them.

Furthermore, Woepcke points out that the Arabs on entering Spain (711 A.D.) would naturally have followed their custom of adopting for the computation of taxes the numerical systems of the countries they conquered, so that the numerals brought from Spain to Italy, not having undergone the same modifications as those of the Eastern Arab empire, would have differed, as they certainly did, from those that came through Bagdad.

The theory is that the Hindu system, without the zero, early reached Alexandria (say 450 a.d.), and that the Neo-Pythagorean love for the mysterious and especially for the Oriental led to its use as something bizarre and cabalistic; that it was then passed along the Mediterranean, reaching Boethius in Athens or in Rome, and to the schools of Spain, being discovered in Africa and Spain by the Arabs even before they themselves knew the improved system with the place value.

A recent theory set forth by Bubnov also deserves mention, chiefly because of the seriousness of purpose shown by this well-known writer. Bubnov holds that the forms first found in Europe are derived from ancient symbols used on the abacus, but that the zero is of Hindu origin. This theory does not seem tenable, however, in the light of the evidence already set forth.

Two questions are presented by Woepcke's theory: (1) What was the nature of these Spanish numerals, and how were they made known to Italy? (2) Did Boethius know them?

The Spanish forms of the numerals were called the *huruf al-gobar*, the *gobar* or dust numerals, as distinguished from the *huruf al-jumal* or alphabetic numerals. Probably the latter, under the influence of the Syrians or Jews, were also used by the Arabs. The significance of the term *gobar* is doubtless that these numerals were written on the dust abacus, this plan being distinct from the counter method of representing numbers. It is also worthy of note that Al-Biruni states that the Hindus often performed numerical computations in the sand.

The gobar numerals themselves were first made known to modern scholars by Silvestre de Sacy, who discovered them in an Arabic manuscript from the library of the ancient abbey of St.-Germain-des-Pres. The system has nine characters, but no zero. A dot above a character indicates tens, two dots hundreds, and so on, 5 meaning 50, and 5 meaning 5000.

In form these gobar numerals resemble our own much more closely than the Arab numerals do. They varied more or less, but were substantially as follows.

This theory of the very early introduction of the numerals into Europe fails in several points. In the first place the early Western forms are not known; in the second place some early Eastern forms are like the gobar, as is seen in the third line, where the forms are from a manuscript written at Shiraz about 970 A.D., and in which some western Arabic forms are also used.

The Boethius Question

First, who was Boethius,—*Divus* Boethius as he was called in the Middle Ages? **Anicius Manlius Severinus Boethius** was born at Rome c. 475. He was a member of the distinguished family of the Anicii, which had for some time before his birth been Christian. Early left an orphan, the tradition is that he was taken to Athens at about the age of ten, and that he remained there eighteen years. He married Rusticiana, daughter of the senator Symmachus, and this union of two such powerful families allowed him to move in the highest circles. Standing strictly for the right, and against all iniquity at court, he became the object of hatred on the part of all the unscrupulous element near the throne, and his bold defense of the ex-consul Albinus, unjustly accused of treason, led to his imprisonment at Pavia and his execution in 524.

Not many generations after his death, the period being one in which historical criticism was at its lowest ebb, the church found it profitable to look upon his execution as a martyrdom. He was accordingly looked upon as a saint,[10] his bones were enshrined,[11] and as a natural consequence his books were among the classics in the church schools for a thousand years. It is pathetic, however, to think of the medieval student trying to extract mental nourishment from a work so abstract, so meaningless, so unnecessarily complicated, as the arithmetic of Boethius.

He was looked upon by his contemporaries and immediate successors as a master, for Cassiodorus (c. 490–c. 585 A.D.) says to him: "Through your translations the music of Pythagoras and the astronomy of Ptolemy are read by those of Italy, and the arithmetic of Nicomachus and the geometry of Euclid are known to those of the West." Founder of the medieval scholasticism, distinguishing the trivium and quadrivium, writing the only classics of his time, Gibbon well called him "the last of the Romans whom Cato or Tully could have acknowledged for their countryman."

The second question relating to Boethius is this: Could he possibly have known the Hindu numerals? In view of the relations that will be shown to have existed between the East and the West, there can only be an affirmative answer to this question. The numerals had existed, without the zero, for several centuries; they had been well known in India; there had been a continued interchange of thought between the East and West; and warriors, ambassadors, scholars, and the restless trader, all had gone back and forth, by land or more frequently by sea, between the Mediterranean lands and the centers of Indian commerce and culture. Boethius could very well have learned one or more forms of Hindu numerals from some traveler or merchant.

To justify this statement it is necessary to speak more fully of these relations between the Far East and Europe. It is true that we have no records of the interchange of learning, in any large way, between eastern Asia and central Europe in the century preceding the time of Boethius. But it is one of the mistakes of scholars to believe that they are the sole transmitters of knowledge.

As a matter of fact there is abundant reason for believing that Hindu numerals would naturally have been known to the Arabs, and even along every trade route to the remote west, long before the zero entered to make their place-value possible, and that the characters, the methods of calculating, the improvements that took place from time to time, the zero when it appeared, and the customs as to solving business problems, would all have been made known from generation to generation along these same trade routes from the Orient to the Occident.

It must always be kept in mind that it was to the tradesman and the wandering scholar that the spread of such learning was due, rather than to the school man. Indeed, Avicenna (980–1037 a.d.) in a short biography of himself relates that when his people were living at Bokhara his father sent him to the house of a grocer to learn the Hindu art of reckoning, in which this grocer (oil dealer, possibly) was expert. Leonardo of Pisa, too, had a similar training.

Even in very remote times, before the Hindu numerals were sculptured in the cave of Nana Ghat, there were trade relations between Arabia and India. Indeed, long before the Aryans went to India the great Turanian race had spread its civilization from the Mediterranean to the Indus. At a much later period the Arabs were the intermediaries between Egypt and Syria on the west, and the farther Orient. In the sixth century B.C., Hecataeus, the father of geography, was acquainted not only with the Mediterranean lands but with the countries as far as the Indus.

A century after Scylax, Herodotus showed considerable knowledge of India, speaking of its cotton

[10] Hence the *Divus* in his name.

[11] Thus Dante, speaking of his burial place in the monastery of St. Pietro in Ciel d'Oro, at Pavia, says: "The saintly soul, that shows / The world's deceitfulness, to all who hear him, / Is, with the sight of all the good that is, / Blest there. The limbs, whence it was driven, lie / Down in Cieldauro; and from martyrdom / And exile came it here."—*Paradiso*, Canto X.

and its gold, telling how Sesostris fitted out ships to sail to that country, and mentioning the routes to the east. These routes were generally by the Red Sea, and had been followed by the Phoenicians and the Sabaeans, and later were taken by the Greeks and Romans.

In the fourth century B.C. the West and East came into very close relations. As early as 330, Pytheas of Massilia (Marseilles) had explored as far north as the northern end of the British Isles and the coasts of the German Sea, while Macedon, in close touch with southern France, was also sending her armies under Alexander through Afghanistan as far east as the Punjab. Pliny tells us that Alexander the Great employed surveyors to measure the roads of India; and one of the great highways is described by Megasthenes, who in 295 B.C., as the ambassador of Seleucus, resided at Pataliputra, the present Patna.

The Hindus also learned the art of coining from the Greeks, or possibly from the Chinese, and the stores of Greco-Hindu coins still found in northern India are a constant source of historical information. The Ramayana speaks of merchants traveling in great caravans and embarking by sea for foreign lands. Ceylon traded with Malacca and Siam, and Java was colonized by Hindu traders, so that mercantile knowledge was being spread about the Indies during all the formative period of the numerals.

As to the Arabs, it is a mistake to feel that their activities began with Mohammed. Commerce had always been held in honor by them, and the Qoreish had annually for many generations sent caravans bearing the spices and textiles of Yemen to the shores of the Mediterranean. In the fifth century they traded by sea with India and even with China, and Hira was an emporium for the wares of the East, so that any numeral system of any part of the trading world could hardly have remained isolated.

Long before the warlike activity of the Arabs, Alexandria had become the great market-place of the world. From this center caravans traversed Arabia to Hadramaut, where they met ships from India. Others went north to Damascus, while still others made their way along the southern shores of the Mediterranean. Hindus were found among the merchants who frequented the bazaars of Alexandria, and Brahmins were reported even in Byzantium.

Such is a very brief resume of the evidence showing that the numerals of the Punjab and of other parts of India as well, and indeed those of China and farther Persia, of Ceylon and the Malay peninsula, might well have been known to the merchants of Alexandria, and even to those of any other seaport of the Mediterranean, in the time of Boethius. The Brahmi numerals would not have attracted the attention of scholars, for they had no zero so far as we know, and therefore they were no better and no worse than those of dozens of other systems.

In answer therefore to the second question, Could Boethius have known the Hindu numerals? the reply must be, without the slightest doubt, that he could easily have known them, and that it would have been strange if a man of his inquiring mind did not pick up many curious bits of information of this kind even though he never thought of making use of them.

As to the fourth question, Did Boethius probably know the numerals? It seems to be a fair conclusion, according to our present evidence, that (1) Boethius might very easily have known these numerals without the zero, but, (2) there is no reliable evidence that he did know them. And just as Boethius might have come in contact with them, so any other inquiring mind might have done so either in his time or at any time before they definitely appeared in the tenth century. These centuries, five in number, represented the darkest of the Dark Ages, and even if these numerals were occasionally met and studied, no trace of them would be likely to show itself in the literature of the period. As a matter of fact, it was not until the ninth or tenth century that there is any tangible evidence of their presence in Christendom.

6 The Development of the Numerals Among the Arabs

When the Arabs, carried on the wave of religious enthusiasm, had spread their conquests from the Atlantic to the Indus, they found everywhere the use of the Greek alphabetic numerals. In Syria they adopted the Greek system, in Egypt the Coptic, and in Persia the system used by the Persians. These systems were, however, of little value for mercantile purposes, and the Arabs early saw the advantages of the Hindu system which had reached them through the translations of the astronomical works of Sindhind.

The spread of the numerals among the Arabs was slow. In the East, under the Abbasid caliphs of Bagdad, the Hindu system found ready acceptance among scholars, but the common people and the merchants continued to use their old systems. It was not until the eleventh century that the Hindu numerals became common in Arabic works, and even then they were used chiefly by mathematicians and astronomers.

The earliest known Arabic work containing the Hindu numerals is the arithmetic of Al-Khowarazmi, already mentioned. This work, which bore the title *Algoritmi de numero Indorum*, was translated into Latin, probably by Adelhard of Bath, in the twelfth century. The Arabic original is lost, but several Latin versions exist, and from these we know that Al-Khowarazmi gave the nine Hindu numerals and explained their use in addition, subtraction, multiplication, and division.

The numerals as used by the Arabs underwent certain modifications. The forms used in the Eastern part of the Arab empire, centering in Bagdad, were different from those used in the Western part, centering in Cairo and later in Spain. The Eastern forms were called *Indien* (Indian), while the Western forms were called *gobar* (dust). The gobar forms, which have already been described in connection with the Boethius question, were the ones used in Spain and Northern Africa, and from these the European forms were derived.

Among the Arab writers who used and spread the Hindu numerals may be mentioned Al-Kindi (800–870), Al-Sufi (died 986), Al-Biruni (973–1048), and Al-Hassar (c. 1150). Al-Kindi wrote several works on arithmetic and the use of the Indian method of reckoning. Al-Sufi wrote a large work on arithmetic. Al-Biruni, as already mentioned, was a great traveler and scholar who used the Hindu numerals extensively in his astronomical and mathematical works. Al-Hassar wrote a work on arithmetic in which the second chapter was "On the dust figures."

The Arabic system of numeration, like the Hindu, was based on the principle of place value. The Arabs adopted the zero, which they called *as-sifr* or *sifr*, and used it in the same way as the Hindus. They also adopted the Hindu methods of calculation, including the rules for the fundamental operations and the extraction of roots. These methods were known as *hisab al-hind*, the Indian reckoning, and *hisab al-gobar*, the dust reckoning.

The spread of the numerals among the Arabs was thus a gradual process, extending over several centuries. It was aided by the great intellectual activity that characterized the Arab world from the ninth to the twelfth century, and by the fact that the Arabs were in contact with both the East and the West, serving as intermediaries in the transmission of knowledge.

7 The Definite Introduction of the Numerals into Europe

The definite introduction of the Hindu-Arabic numerals into Europe is usually attributed to Leonardo of Pisa, known as Fibonacci, who wrote his famous *Liber Abbaci* in 1202. Leonardo had traveled extensively in the East, visiting Egypt, Syria, Greece, and Sicily, and had learned the Hindu methods of reckoning from Arab teachers. In his book he explained the use of the nine Indian figures and the zero, and showed how calculations could be performed with them.

The *Liber Abbaci* was not, however, the first work in Europe to mention the Hindu numerals. As we have seen, there is evidence that the numerals were known in Spain and possibly in other parts of Europe as early as the tenth century. But Leonardo's work was the first to give a systematic and comprehensive treatment of the new system, and it was through his influence that the numerals were gradually adopted in Italy and thence spread to other parts of Europe.

The reception of the numerals in Europe was slow. The medieval universities, with their devotion to the classics, were at first hostile to the new system, which was considered a novelty of doubtful value. The Roman numerals were deeply entrenched in European culture, and it required several centuries before the Hindu-Arabic numerals were generally accepted.

An important figure in the early history of the numerals in Europe was Gerbert, who became Pope Sylvester II in 999. Gerbert had studied in Spain and had learned the numerals there. He introduced them into the schools of France, but in a modified form, using apices or counters on an abacus rather than the written numerals. This system, known as the Gerbert abacus, was used in European schools for several centuries.

The real progress in the adoption of the numerals came in the thirteenth and fourteenth centuries, when Italian merchants began to use them in their business transactions. The commercial needs of the Italian city-states, with their extensive trade with the East, made the new system highly attractive. The Italian mathematicians, following Leonardo of Pisa, wrote numerous works on arithmetic in which the Hindu numerals were used and explained.

Among the early European works on arithmetic using the Hindu numerals may be mentioned the *Carmen de Algorismo* of Alexander de Villa Dei (c. 1240), the *Algorismus* of Johannes de Sacrobosco (c. 1250), and the various Italian arithmetics of the fourteenth century. These works helped to spread the knowledge of the numerals and to familiarize European scholars and merchants with the new system.

The spread of the numerals from Italy to the rest of Europe was a gradual process. France and Germany adopted them in the fifteenth century, and England in the sixteenth. Even then, the Roman numerals continued to be used for many purposes, and it was not until the eighteenth century that the Hindu-Arabic numerals became completely dominant in Europe.

8 The Spread of the Numerals in Europe

After the numerals had been introduced into Italy, they gradually spread to the other countries of Europe. The process was slow, for the Roman numerals were deeply rooted in European culture, and the new system encountered strong resistance from conservative scholars and ecclesiastics.

In France, the Hindu numerals were known as early as the thirteenth century, but they did not come into general use until the fifteenth. The first French work to use them extensively was the arithmetic of Jehan Adam, written in 1475. In the sixteenth century, French mathematicians such as Chuquet (1484) and Trenchant (1566) helped to popularize the new system.

In Germany, the numerals were introduced in the fifteenth century, but they met with strong opposition. The German merchants preferred to use the abacus, and the scholars were attached to the Roman system. It was not until the publication of the arithmetic of Kobel in 1514 that the Hindu numerals began to gain ground in Germany.

In England, the numerals were known in the fourteenth century, but they came into general use only in the seventeenth century. The first English work on arithmetic, *The Ground of Artes* by Robert Recorde (1542), used the Hindu numerals, but it was long before they replaced the Roman system entirely.

In Spain and Portugal, where the numerals had been known since the Moorish conquest, they were adopted earlier than in the rest of Europe. But even there the transition was gradual, and the Roman numerals continued to be used for official and ecclesiastical purposes for many centuries.

The spread of the numerals in Europe was thus a slow and gradual process, extending over several centuries. It was aided by the invention of printing, which made it possible to publish arithmetic textbooks in large numbers, and by the growth of commerce, which created a demand for a more efficient system of numeration. By the eighteenth century, the Hindu-Arabic numerals had become the standard system throughout Europe, and they have since spread to all parts of the world.

Conclusion

In conclusion, we may say that the history of the Hindu-Arabic numerals is a long and complex one, stretching over many centuries and involving many different civilizations. From their earliest beginnings in India, through their development by the Arabs, to their final adoption in Europe, these numerals have had a remarkable career. They are one of the most important inventions in the history of civilization, and their influence on science, commerce, and daily life can hardly be overestimated.

The study of their history reveals the deep connections that have always existed between the peoples of the East and the West, and shows how knowledge has been transmitted from one civilization to another along the great trade routes that have linked the world since ancient times. It is a story that deserves to be known by all who use these symbols, which have become so familiar that we are apt to forget how much labor and ingenuity went into their creation.

End of Part I

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